

Arpitsinh Vaghela

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Education

Arizona State University

Aug 2024 – Ongoing

PhD. in Computer Science

- **Advisor:** Dr. Yezhou Yang
- **Research Interest:** Generative Modeling, 3D Scene Representation, 3D Gaussian Splatting.

Arizona State University

Aug 2022 – May 2024

Masters of Science in Computer Science

- GPA: 3.78/4.0
- **Thesis:** “Multi-Camera Bird-Eye-View Occupancy Detection for Intelligent Transportation System” under the guidance of Dr. Yezhou Yang

Ahmedabad University

Aug 2018 – May 2022

Bachelors of Technology in Information and Communication Technology

- GPA: 3.30/4.0
- **Thesis:** *Exploring Generalization Capabilities of Dynamic Neural Expansion* under the guidance of Dr. Mehul Raval

Experience

Research Assistant

Tempe, AZ

Arizona State University

Aug 2024 – Ongoing

Student Developer

Remote

Google Summer of Code — OpenMined

May 2021 – Aug 2021

Full Stack Developer Intern

Ahmedabad, India

PedalsUp

May 2020 – July 2020

Ongoing Research

Inversion-Free Image Editing in Discrete Diffusion Models

Exploring efficient image editing through modifications to the generative process in discrete diffusion models without explicit inversion.

Controllable Scenario Generation for Autonomous Driving

Developing a framework for controllable traffic scenario generation by combining scenario-level editing with Gaussian splatting, addressing novel view rendering artifacts for traffic agents with sparse-view data.

Publications

<https://scholar.google.com/citations?user=OhpRnmsAAAA>

1. **Arpitsinh Vaghela**, Aayush Atul Verma, Kaustav Chanda, Chi-Yao Huang, Duo Lu, Bharatesh Chakravarthi, Yezhou Yang “Event-Guided Dynamic Reconstruction under Extreme Motion Blur,” in The 1st Workshop on Physical World AI, (NeurIPS), 2026.
2. Aayush Atul Verma, **Arpitsinh Vaghela**, Bharatesh Chakravarthi, Kaustav Chanda, and Yezhou Yang “Event-based Graph Representation with Spatial and Motion Vectors for Asynchronous Object Detection” in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.
3. Kaustav Chanda, Aayush Atul Verma, **Arpitsinh Vaghela**, Yezhou Yang, and Bharatesh Chakravarthi, “SE-Pose: A Synthetic Event-based Human Pose Estimation Dataset for Pedestrian Monitoring,” in *Proceedings of the IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
4. Kaustav Chanda, Aayush Atul Verma, **Arpitsinh Vaghela**, Yezhou Yang, and Bharatesh Chakravarthi, “Event quality score (eqs): Assessing the realism of simulated event camera streams via distance in latent

space,” in *CVPR 2025 Fifth International Workshop on Event-Based Vision*, 2025.

5. **Arpitsinh Vaghela**, Duo Lu, Aayush Atul Verma, Bharatesh Chakravarthi, Hua Wei, and Yezhou Yang, “MC-BEVRO: Multi-Camera Bird Eye View Road Occupancy Detection for Traffic Monitoring,” in *Proceedings of the IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
6. Tiejun Chen, Prithvi Shirke, Bharatesh Chakravarthi, **Arpitsinh Vaghela**, Longchao Da, Duo Lu, Yezhou Yang, and Hua Wei, “SynTraC: A Synthetic Dataset for Traffic Signal Control from Traffic Monitoring Cameras,” in *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, 2024.
7. Aayush Atul Verma, Bharatesh Chakravarthi, **Arpitsinh Vaghela**, Hua Wei, and Yezhou Yang, “etram: Event-based traffic monitoring dataset,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
8. Manideep Reddy Aliminati, Bharatesh Chakravarthi, Aayush Atul Verma, **Arpitsinh Vaghela**, Hua Wei, Xuesong Zhou, and Yezhou Yang, “Sevd: Synthetic event-based vision dataset for ego and fixed traffic perception,” in *CVPR 2024 Workshop on Synthetic Data for Computer Vision*, 2024.

Teaching Experience

Teaching Assistant <i>Arizona State University</i>	<i>Fall 2025</i>
◦ CSE 434 <i>Computer Networks</i> (Instructor: Dr. Bharatesh Chakravarthi).	
Teaching Assistant <i>Arizona State University</i>	<i>Spring 2025</i>
◦ CSE 434 <i>Computer Networks</i> (Instructor: Dr. Bharatesh Chakravarthi).	
Grader <i>Arizona State University</i>	<i>Fall 2023</i>
◦ CSE 512 <i>Distributed Database System</i> (Instructor: Dr. Bharatesh Chakravarthi).	
Teaching Assistant <i>Ahmedabad University</i>	<i>Spring 2022</i>
◦ <i>Advanced Statistics and Computer Networks</i> (Instructor: Dr. Shashi Prabh).	

Academic Service

- **Organizer**, *EVGEN 2026: The 2nd Workshop On Event-Based Vision in The Era of Generative AI - Transforming Perception And Visual Innovation*, Workshop, WACV 2026, Tucson, United States.
- **Organizer**, *EVGEN 2025: Event-Based Vision in the Era of Generative AI – Transforming Perception and Visual Innovation*, Workshop, WACV 2025, Tucson, United States.
- **Reviewer**, *WACV 2027, NeurIPS 2026, IROS 2025, ITSC 2025, WACV 2025, NeurIPS Workshop RBFM 2024*.